ECE250: Signals & Systems Quiz-1

Date: 27-Aug-2025 **Total Points:** 10 **Duration:** 15 Minutes

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Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

[CO1] **[Q1] [5 Points]**: Determine whether or not the below signals are periodic. If periodic, determine the fundamental time period.

(a)

$$x(t) = e^{j(\pi t - 1)}$$

(b)

$$x[n] = cos\left(\frac{\pi n^2}{8}\right)$$

[CO1] **[Q2] [5 Points]**: Consider the below discrete-time signal:

$$x[n] = -1 + \sum_{k=2}^{\infty} \delta[n-k]$$

(a) Plot this signal.
(b) Determine the values of constants A, M and n_0, so that $x[n]$ may be expressed as $x[n] = Au[Mn+n_0]$.